Week 7 Lecture 1

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Materials	
Module #	38
Type	Lecture
Week #	7

Memory Hierarchy

Types of Storage Elements

- On-chip registers: 10s-100s of bytes
- SRAM (Cache): 0.1-1MB # Static RAM
- DRAM (Memory/RAM of a PC): 0.1-16GB
- Solid-State Disk (SSD) - Flash: 1-128GB
- Magnetic Disk (HDD-Hard Disk Drive): 0.1-10TB

Storage Parameters

- **Latency:** Time to read the first value from a storage location (lower is better)
    - Register < SRAM < DRAM < SSD < HDD
    - So, the turnaround time when trying to read something from a ...
        - Register: in order of nanoseconds
        - SRAM: tens to hundreds of nanoseconds
        - DRAM: several microseconds
        - SSDs: hundreds of microseconds
        - HDDS: order of milliseconds
- **Throughput:** number of bytes/second that can be read (higher is better)
    - DRAM > SSD > HDD (registers, SRAM have limited capacity)

- **Density:** Number of bits stored per unit area/cost (higher is better)
  - Volume manufacture important
  - HDD > SSD > DRAM > SRAM > Registers

Computer Organization

- CPU has as many registers as possible
- Backed by L1, L2, L3 cache (SRAM)
- Backed by several Gigabytes (GBs) of DRAM working memory
- Backed by SSD for high throughput
- Backed by HDD for high capacity
- Backed by long-term storage, backup

Cold Storage

- Backups & archives:
  - Huge amounts of data
  - Not read very often
  - Can tolerate high read latency
- Amazon Glacier, Google, Azure Cold/Archive storage classes
- High latency of retrieval → up to 48 hours
- Very high durability
- Very low cost

Impact on Application Development

- Plan the storage needs based on the application growth
- Speed of app determined by types of data stored, how stored
- Some data stores are more efficient for some types of read/write operations

Developer must be aware of the choices and what kind of DB to choose for a given application